

## Radical equations



1

a  $x = 13$

b  $h = 25$

c  $x = \frac{9}{64}$

d  $k = 16$

2 A radical equation is an equation where the variable is inside a radical.

3 No

4 No

5 Yes

6

a  $y = 49$

b  $x = 25$

c  $p = 9$

d  $y = \frac{9}{25}$

e  $x = \frac{49}{144}$

f  $x = 64$

g  $y = 4$

h  $c = 9$

i  $b = 25$

j  $a = 9$

k  $b = \frac{1}{25}$

l  $a = 4$

7

a  $x = 30$

b  $x = 5$

c  $x = 54$

d  $x = 11, -3$

e  $x = 4, 2$

f  $x = 18$

g  $x = 16$

h  $x = 22$

i  $x = 135$

j  $x = 121$

k  $x = 4, -4$

8

a No

b Ray did not know that  $\sqrt{x+3}$  must always be non-negative. So at step 2, he should have known there was no solution.

9

a Tom did not divide every term by 2 in step 2, so he should have also divided  $-10$  by 2.

b In step 4 Katrina took the square root of 9, when she should have squared it.

c  $x = 10$

10 Example answers:



- In order for the left-hand side to equal zero, the square root of the number would have to be negative.
- The left-hand side is greater than zero for all values of  $x$ .

11

a  $x = \frac{1}{4}, \frac{1}{25}$

b  $x = \frac{1}{25}$

12

a  $x = 9$

b No

13 Raise each side of the equation to the power  $\frac{5}{3}$ .

14

a  $x = 57$

b  $x = 4$

c  $x = 79$

15

a  $x = 9$

b  $x = \pm 128$

c  $x = 81$

d  $x = \pm 128$

## Applications

16

a

|     |     |      |       |
|-----|-----|------|-------|
| $V$ | 40  | 400  | 4 000 |
| $b$ | 3.2 | 10.0 | 31.6  |

b No

17

a  $t = \sqrt{4-x} + 4$

b  $t = 6$

c 0.2 seconds

18

a 8

b Fresh gale

c  $s = 115$

d  $55 \leq s \leq 64$